



The ecology and biodiversity of urban ponds

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Recent research has demonstrated that ponds contribute a great deal to biodiversity at a regional level as networks of habitat patches that also act as 'stepping stones' to facilitate the movement of species through the landscape. Similarly, a great deal of biodiversity persists in urban environments where synanthropic communities are supplemented by species that thrive in disturbed environments. Aquatic urban biodiversity appears to persist despite anthropogenic stressors: an array of anthropogenic pollutants (road salt and heavy metals), invasive species, and active mismanagement—particularly the removal of riparian vegetation. Optimizing urban ponds for different ecosystem services results in conflicting priorities over hydrological, geochemical, ecological, aesthetic, and cultural functions. The socio-ecosystem approach to environmental management opens a path to greater incorporation of biodiversity into town planning and sustainability, while accounting for cultural attitudes to urban ecosystems. I identify a range of research needs: (1) the roles of design and location of urban ponds in influencing biodiversity, (2) the function of urban wetlands for stormwater and pollution management, and (3) public perceptions of urban ecosystems and how those perceptions are influenced by interactions with natural systems. Urban wetlands offer an important opportunity to educate the general public on natural systems and science in general using a resource that is located on their doorstep. In the face of increasing pressures on natural systems and increasing extent and intensity of urbanization, a more comprehensive appreciation of the challenges and opportunities provided by urban ponds could play a substantial role in driving sustainable urban development. © 2014 Wiley Periodicals, Inc.

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INTRODUCTION

Land-use change, whether a conversion from natural habitat to agricultural or urban land, is likely to be the principle driver of biodiversity declines over the next century in all biomes.¹ Current projections of urban land use suggest that between 2000 and 2030 there will be at least a 185% increase in the extent of urban areas² (Figure 1), posing a serious threat to biodiversity around the world, and much of this threat is concentrated in high biodiversity areas in developing countries.³ However, concomitant plans for urban intensification

in developed countries bring a parallel set of problems through a reduction in remaining habitat patches through processes such as infill housing.^{4,5} When attempting to mitigate the environmental consequences of this rapid expansion of towns and cities, it is important that the creation of these urban areas not be thought of simply as the removal of natural habitat. The processes that drive urbanization involve complex, interacting sets of physical, social, economic, and governmental institutions with complex sets of interacting stakeholders.⁶ With increasing demands being placed upon the natural world, it is important to consider this range of institutions when attempting to safeguard biodiversity in the long term. Furthermore, regional variations in sociopolitical priorities necessitate local approaches to the management of this problem.

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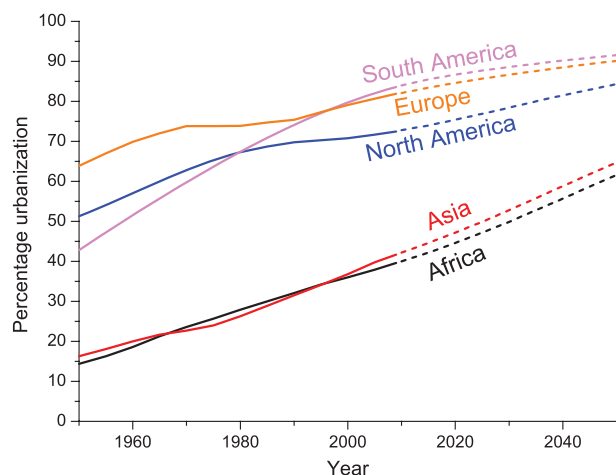


FIGURE 1 | Past trends and future projections of urbanization (% population living in urban areas) by continent. Source: Population Division of the Department of Economic and Social Affairs of the United Nations Secretariat World Population Prospects: The 2008 Revision and World Urbanization Prospects: The 2009 Revision. <http://esa.un.org/wup2009/unup/>.

Approaches to the protection of biodiversity in the face of urbanization require interdisciplinary collaboration with researchers and practitioners in a range of other fields, including urban planners, economists, and sociologists, to provide a broader perspective on the ‘socio-ecosystem’.^{7,8} Indeed, successful interdisciplinary approaches to the protection and enhancement of biodiversity under urbanization could not only offset the negative impacts on biodiversity but also facilitate a more rapid transition to sustainability.⁶

Freshwaters represent a set of habitats that suffer greater biodiversity declines than terrestrial habitats,¹ perhaps owing to the disproportionate biodiversity that is found in inland waters.⁹ Threats to these habitats tend to result from five key factors: species invasion, habitat degradation, water pollution, overexploitation, and flow modification.⁹ The remainder of this article will consider the topic of urban pond ecology from two opposite angles: after providing an overview of the ecology of ponds and the nature of urbanization, I shall first discuss the positive and negative impacts that urbanization has on pond ecosystems. This will cover topics such as pollution, habitat connectivity, and neglect, but also pond creation for amenity. Second, I shall provide an overview of the contributions made by ponds to ecosystem services within urban areas. In particular, I will emphasize the conflict between competing interests in limited urban spaces, but in closing I will summarize some of the many promising avenues for the protection, use, and development of this

habitat. The review will focus predominantly on the literature from northwest Europe, where the majority of work has been carried out, with notes about future directions in other regions.

THE VALUE OF URBAN PONDS

Biodiversity

Pond Ecosystems

Before giving closer consideration to ponds in urban areas, it is useful to understand the nature of small, lentic water bodies in general. The definition of a ‘pond’ is an artificial one that varies between researchers. While a wide range of potential definitions exist, ponds are generally defined in terms of their area: being either $<2 \text{ ha}^{10}$ or $<5 \text{ ha}^{11}$. Small landscape elements such as ponds have traditionally been considered as providing insignificant biodiversity to the regional species pool compared with larger habitats such as lakes and rivers.¹² However, while many individual ponds may contain relatively few species (α -diversity), these habitats constitute an enormous diversity of abiotic and biotic conditions. This diversity of environments creates a concomitant diversity in ecological communities (β -diversity), which, in turn, results in a greater contribution to landscape-level biodiversity (γ -diversity) than those of larger wetlands that are more homogeneous.^{13–15} In addition to this complexity, the small size of ponds is thought to break down standard species–area relationship owing to the small island effect.^{16,17} This stochasticity means that a pond that holds a high biodiversity at one time point may not remain of high ecological value at another,¹⁸ rendering site-specific conservation measures ineffective and instead necessitating the conservation of pond clusters or networks.¹⁹

Ponds have also been overlooked from a legislative standpoint, being omitted from the EU Water Framework Directive (2000/60/EC) that dictates standards for water quality.¹² While monitoring of lakes and rivers occurs in the EU and worldwide to ensure compliance with environmental legislation such as the WFD, ponds are not monitored. Certain standing waters are protected under EU legislation such as the EU Habitats Directive (92/43/EEC) Annex I, including dystrophic lakes/ponds and Mediterranean temporary ponds, and others can be protected based on floral or faunal communities. In some countries, such as the UK, high-quality ponds have been recognized as priority habitats, and therefore receive some statutory protection. However, an absence of monitoring of

sites may lead to conservationists failing to recognize such sites.

Urban Ecosystems

Because the urban environment is tailored to human needs, urban areas share many features in common irrespective of geographical proximity²⁰ and are influenced by the same network of processes.²¹ This fact, combined with the unique socioeconomic and cultural interactions between urban habitats and human populations has led to a call for a discrete field of 'urban ecology' to be founded.²² It is often considered that by creating such uniform environmental conditions, urbanization homogenizes biological communities.²³ This 'biotic homogenization'²⁴ occurs through three complementary processes of (1) exclusion of native species through habitat modification, (2) the introduction of exotic species through human processes (explored in more detail below), and (3) the establishment of exotic species through habitat disturbance. However, the details of these processes remain unclear.²⁵ Based on published floral inventories for 54 Central European cities, Pyšek²⁶ found that an average of 40% of urban floral communities comprised alien species (range: 20–60%). It has been argued that this modification of floral communities is the only direct biological modification made by humans, and that faunal responses are determined by this plant 'template'.²⁷ Similar ratios occur in introduced versus native bird communities.²⁸ The net result of urbanization is not always a decline in species richness: studies comparing varying levels of urbanization show that while invertebrates and birds exhibit considerable monotonic declines (though cf. Ref 29 with respect to birds) with increasing urbanization, plant species richness peaks at intermediate levels.³⁰ Furthermore, trends seem to vary markedly between studies³⁰ and even between rural–urban transects in the same region.³¹

Urban Ponds

In a review of anthropogenic refuges for freshwater biodiversity, Chester and Robson describe 16 types of man-made freshwaters of which 'urban pond' is a single category.³² However, urban ponds are a diverse group of habitats that vary in their characteristics, and in Table 1 I have proposed a typology of these urban ponds in terms of their primary function: garden pond, industrial ponds, ornamental lakes, drainage systems, and nature reserves. Note that this table is by no means comprehensive. I have omitted unusual (though fascinating) systems such as bomb crater ponds, e.g.,

Refs 49 and 50, swimming pools, e.g., Ref 51, and monumental fountains, e.g., Ref 52, in favor of those habitats that are more common and better-studied. Note that while some other 'unusual' habitats (such as stormwater management facilities) are very well studied, ponds dedicated to the preservation of nature in urban areas are less well known. This leaves open the question of whether urban nature reserves either contain a large number of urban species or represent a non-urban, 'natural' community within an urban matrix. Further, it is important to note that the typology is not static: it is not uncommon for water bodies to change functions, such as the adoption of industrial ponds by angling clubs.³⁶ While this management can reduce diversity, it also reduces the likelihood of the water body being lost owing to development or drainage.^{36,37} Such studies of the fate of urban wetlands under demographic and economic transitions are rare, but will become important as developing countries move away from industrial and manufacturing economies toward the service industry.

The extent of biodiversity contained within urban ponds varies markedly in terms of extent and composition. While a range of studies have reported (with some surprise) that urban wetlands can support substantial biodiversity despite being in close proximity to human habitats,^{38,41,53,54} it is unclear as to whether this is due to the lack of reporting of poor-quality urban wetlands that are considered uninteresting. Table 2 gives a summary of studies that have been conducted involving the measurement of biodiversity in urban ponds. Biodiversity of certain groups has received more attention than that of others, and amphibians have been particularly well studied. Amphibians appear to follow the general trend of a decline in diversity and abundance toward the center of built-up areas,⁶⁹ which is likely due to a combination of low habitat quality (in particular, ornamental edging made from stone or wood reduces amphibian diversity owing to amphibians not being able to climb the vertical surface) and poor connectivity between habitat patches.³⁹ However, it is important to consider species-specific sensitivity, as some species appear to be quite resilient to the effects of urbanization,⁷⁰ and so declines in diversity may represent the loss of particular, disproportionately affected species rather than a uniform effect on the entire species pool.

Fish diversity is rarely considered within urban ponds, apart from in the contexts of (1) introductions of alien species by residents,⁵⁸ or (2) as a presence/absence of variable influencing the composition of macroinvertebrate communities.⁷¹ While the low dispersal ability of fish species through

TABLE 1 | A Proposed Typology of Urban Ponds Based on Primary Function

Urban Pond Type	Characteristics	Refs
Garden pond	• Small size (10^1 m²) • Set within an impervious matrix • Often stocked with fish • Very rarely dry out • Maintained to prevent succession	33–35
Industrial ponds	• Medium size (10^2–10^4 m²) • Urban or periurban, often away from residential areas • Sometimes contaminated • Constructed to hold water for use, or left after mineral extraction • Rarely in use for original purpose	36–38
Ornamental lakes and ponds	• Medium-large size (10^2–10^6 m²) • Heavily managed for aesthetics • Hypertrophic • Fish and ducks encouraged and fed • Access to public is encouraged and uncontrolled • Often with vertical sides that may pose problems for animals	39,40
Drainage systems	• Highly variable in size (10^2–10^6 m²) • Primary function is hydrological management • Diverse design • Wide variation in 'naturalness' • Temporary ('detention basins') or permanent ('retention ponds')	41–48
Nature reserves	• Medium-large size (10^2–10^6 m²) • Managed primarily for biodiversity (often birds) • Either co-opted natural ponds or created to appear natural • Access to public is encouraged but controlled	

terrestrial matrices, particularly in urban areas, likely reduces the incidence of natural ecological processes of colonization, extinction, and community assembly, urban fish populations require greater study as they are key drivers of ecosystem functioning. Similarly, urban aquatic plant communities tend to be viewed as anthropogenic imports rather than embattled native communities (more on invasive plants below). One exception is planktonic communities, which have received particular attention because of the potential for nuisance species to become established periodically in disturbed and temporary wetlands.^{72,73}

Insect biodiversity has also been measured, although to a lesser extent. Urban ponds in Germany have been shown to have the potential to contain a large number of dragonfly and damselfly (Odonata) species, although diversity is strongly related to the presence of vegetation.⁴⁰ Urban water bodies can house a considerable portion of the national species pool for some invertebrate taxa (e.g., Hirudinea,

Gastropoda, and Tricladida), but more sensitive taxa (e.g., Plecoptera and Ephemeroptera) tend to be excluded.⁵³ However, the converse can also be true: in natural areas that are influenced by human activity, certain artificial water bodies such as reservoirs can act as refugia for species.^{41,42,61}

A brief consideration of the range of biodiversity studies reveals two significant patterns. The first is that there have been no comprehensive food web studies looking at urban environments. This step will be important because of the unique combinations of anthropogenic factors that are influencing these habitats and the health consequences for humans living in the surrounding terrestrial matrix in terms of pathogens, disease vectors, and pollutants. Indeed, there is a general lack of comparable studies that might enable us to test for the homogenization of urban freshwater ecosystems.⁷⁴ While we generally consider urban ponds to be 'ponds in urban environments', it could be that

TABLE 2 | Summary of a Range of Biodiversity Studies Conducted on Urban Ponds

Groups Studied	Water Chemistry	Location	N	Types of Ponds	Finding	Ref
Amphibians (6 sp)	None	Maryland, USA	53	16 stormwater ponds, 16 artificial ponds, and 21 natural ponds	Artificial ponds are valuable breeding habitat for amphibians, especially when natural wetlands are scarce	41
Amphibians (7 sp)	pH and conductivity	New Jersey, USA	39	Stormwater ponds	Connectivity and fish presence determine the presence and composition of amphibian communities	45
Amphibians (9 sp)	Conductivity	Australia	30	Stormwater ponds	Nine frog species were found, which responded differently to disturbance, vegetation, and connectivity	43
Amphibians (6 sp)	Conductivity	Australia	65	Stormwater, ornamental, and other constructed	Amphibian diversity was higher with greater surrounding green space and lower with high human population density and water conductivity	47
Amphibians (9 sp)	None	Australia	104	Park and garden ponds in urban and rural areas	Urban amphibians exist in metacommunities, with diversity related to isolation and size of ponds and the presence of vertical walls	39
Amphibians (6 sp)	pH	Canada	29	Urban, agricultural, and forested ponds	Amphibian diversity and abundance was lower in urban areas	55
Amphibians (12 sp)	None	North Carolina, USA	25	Stormwater ponds	Anuran presence decreased with increasing distance to the riparian zone, and pond age had a range of effects on different species	56
Autecological study of <i>Rana temporaria</i>	None	UK	13	Garden, park, and rural comparison	Urban areas constitute barriers to gene flow in amphibians	57
Autecological study of <i>Nerodia clarkii compressicauda</i>	None	Florida, USA	2	Stormwater retention ponds	Stormwater ponds harbored exceptionally high biomass of snakes until treatment with glyphosate	48
Fish (11 sp, several varieties)	None	UK	18	Urban park ponds (range of origins)	Introductions of non-native fish increase with public access to a pond	58
Fish (4 sp)	Dissolved oxygen, pH, alkalinity, turbidity, conductivity, chlorides, nitrogen, total dissolved solids, volatile solids, chlorophyll <i>a</i> , iron, and sulfates	Illinois, USA	1	Stormwater ponds	Stormwater retention ponds can provide a habitat for native, endangered fish	42

TABLE 2 | Continued

Groups Studied	Water Chemistry	Location	N	Types of Ponds	Finding	Ref
Water birds (39 sp)	None	France	11	Gravel pits	Eleven urban gravel pits contain more than half the regional species pool of water birds	38
Invertebrates (56 fam)	pH, conductivity, dissolved oxygen, nitrate, and phosphate	UK	36	Industrial	Former industrial ponds provide refuge for invertebrates in urban areas. Conversion to angling ponds reduces diversity but decreases the probability of pond loss through drainage	36,37
Aquatic Hemiptera (26 sp)	Specific conductance, total dissolved solids, salinity, and dissolved oxygen	Wisconsin, USA	28	Stormwater ponds	Pond shape, land cover, and fish abundance impact Hemiptera communities	46
Odonata (30 sp)	None	Germany	33	Range from forest/bushland, agricultural land, residential/commercial areas, and city parks	Different odonate assemblages are associated with different pond types, and at least one species thrives in urban areas	40
Coleoptera (40 sp) and Hemiptera (17 sp)	Conductivity, dissolved oxygen, and pH	South Africa	18	Urban and periurban irrigation reservoirs and attenuation ponds	Artificial ponds provide valuable habitat for insects in a biodiversity hotspot	59
Insects (32 sp)	pH, hardness, conductivity, chloride, total phosphorus, ammonium, and dissolved oxygen	Argentina	4	Natural and artificial ponds in city parks	Removal of aquatic vegetation during management influences aquatic insect communities	60
Molluscs (21 sp)	pH and calcium	Singapore	24	Reservoirs, park ponds, canals, and streams	Reservoirs are an important refuge for molluscs in tropical urban areas, and also harbor invasive species	61
Cladocera (26 sp)	Total phosphorus	Canada	18	Temporary ponds, permanent lakes, and wetlands	Cladoceran communities varied between temporary ponds, permanent lakes, and wetlands, but all contributed to γ diversity	62

TABLE 2 | Continued

Groups Studied	Water Chemistry	Location	N	Types of Ponds	Finding	Ref
Testate amoebae (49 sp)	pH, electrolytic conductivity, As, Pb, Cd, Mn, total organic carbon, total nitrogen, inorganic nitrogen, phosphorus, dissolved oxygen, oxygen saturation, and polyaromatic hydrocarbons	Poland	4	Stormwater ponds	Negative effect of mineral ions and variable effects of nutrients and temperature on testate amoebae, strong seasonal variations	63
Rotifers (114 sp)	Total phosphorus and total nitrogen	Poland	19	Natural and artificial ponds, clay pits, and pools	Urban ponds contained 25% of total Polish rotifer species pool, with strong variation in assemblages between the diverse array of sites	54
Macrophytes (49 sp)	pH, dissolved oxygen, total nitrogen, total phosphorus, dissolved organic carbon, suspended solids, and chlorophyll <i>a</i>	Japan	55	Range of ponds selected on the basis of land cover, including urban	Urban land cover and pond enlargement reduced macrophyte diversity	64
Macrophytes (57 sp), invertebrates (119 sp), and amphibians (4 sp)	pH, conductivity, dissolved oxygen, and hardness	UK	37	Not stated, range of successional states and ages	Connectivity is the primary driver of urban pond biodiversity	65
Macrophytes (> 50 sp) and invertebrates (31 fam)	None	North Carolina, USA	20	10 Constructed stormwater wetlands (CSWs) and 10 artificial urban ponds	Ponds and CSW have similar invertebrate diversity but different community structure	44
Macrophytes (3 sp), zooplankton (19 sp), molluscs (3 sp), amphibians (8 sp), and fish (6 sp)	Chlorophyll <i>a</i> , total nitrogen, total phosphorus, total dissolved nitrogen, total dissolved phosphorus, dissolved nitrate, specific conductance, % oxygen saturation, chloride, sulfate, and total dissolved solids	Wisconsin, USA	23	Artificial ponds	Water chemistry, pond morphology, and land cover (particularly % cover of lawns and meadows) correlated with diversity of different components of the biota	66

TABLE 2 | Continued

Groups Studied	Water Chemistry	Location	N	Types of Ponds	Finding	Ref
Invertebrates (7 ord) and amphibians (1 sp)	None	UK	19	Experimental garden ponds	Colonization of small garden ponds over 23 months mostly by Diptera, with other invertebrates infrequent. Ponds were used by amphibians, although no breeding was recorded	34
Cladocera (16 gen), fish (16 sp), rotifers*, copepods*, and phytoplankton*	Chlorophyll a, total phosphorus, soluble reactive phosphorus, and dissolved inorganic nitrogen	Belgium	13	Overflow and flow-through ponds in Forest and park artificial ponds	Fish recolonization after biomanipulation to restore clear-water states in ponds affects zooplankton communities, and this is mediated by submerged macrophyte cover	67,68

sp, species; gen, genera; fam, families; ord, orders; *, diversity not stated.

we need to consider these habitats as no-analog, qualitatively different ecosystems from those ponds that exist in other landscape contexts. Also, different anthropogenic pressures act on different components of the ecosystem: invasive fish act as top predators, eutrophication influences primary production, and habitat isolation acts on macroinvertebrates and amphibians. Second, there are a number of species that persist despite intensive human activity. Examples include the damselfly *Ischnura elegans*, which tolerates and thrives in urban areas even when other Odonata are highly sensitive to pollution and pond morphology,⁴⁰ the goldfish *Carassius auratus*, which is also tolerant of degraded water bodies,⁷⁴ and the mallard *Anas platyrhynchos*, urban populations of which show genetic differentiation from nearby rural populations.⁷⁵ In the past these eurytopic species have been considered as generalists that can survive a range of conditions, and attention has been focused on why excluded species are unable to persist in urban environments. However, a greater consideration of the drivers of ‘commonness’ is called for. In particular, where biodiversity is reduced through the exclusion of a portion of the species pool, this magnifies the ecological role of those common species that remain.⁷⁶ What are the traits that enable species to occur in a wide range of environments and what are the consequences of declines in the abundance of these common species for biodiversity and ecosystem function?

Function

Ecological Function

Ponds act as more than just containers of biodiversity, providing a range of ecosystem services.⁷⁷ Ponds constitute a network of distributed, discrete habitat patches sometimes referred to as the ‘pondscape’.⁷⁸ This network can function in two ways depending on the focal species. For amphibians, for example, ponds can function as ‘stepping stones’ across a matrix of inhospitable terrain⁷⁹ where individual ponds might be unsuitable for breeding but are vital temporary habitats during onward dispersal across the terrestrial matrix.^{80–82} This function in particular was highlighted by the European Habitats Directive (98/83/EC), which points out that ‘... stepping stones (such as ponds or small woods), are essential for the migration, dispersal and genetic exchange of wild species’. For other species, the discrete nature of the aquatic environment within a terrestrial matrix creates a classical metapopulation situation,⁸³ where most or all ponds are suitable habitat and dispersal occurs between generations. The introduction of the

Natura2000 network across EU member states has further emphasized the need to consider ecological coherence in conservation planning, providing a strong motivation for the creation and protection of habitats such as ponds. Complementing existing network corridors by reducing the barrier effects of urban areas can be accomplished by enhancing habitat connectivity within those areas, for example, through the creation and maintenance of a high-quality urban pondscape.

Water Management and Treatment

Within an urban context, ponds are frequently used to control stormwater flow.⁸⁴ This practice (called ‘sustainable urban drainage’) is mandatory under building regulations in a number of countries and results in heavily managed wetlands.⁸⁵ Studies of these wetlands have shown that while there is considerable variation in community structure, these water bodies can provide a surprising amount of diversity to a regional species pool,⁵³ particularly when other habitats are rare.^{43,44} This run-off from human-modified land brings with it heavy metals and nutrients that can be retained by ponds^{86,87} (Box 1). The retention of nutrients within agricultural landscapes leads to extremely high productivity and associated levels of carbon sequestration that exceed those in any other habitat on the planet.⁸⁸

BOX 1

BIOREMEDIATION USING WETLAND BIOTA

Among the many ecosystem services provided by urban water bodies, bioremediation has been identified as a plausible alternative to chemical filtering to remove heavy metals.⁹² Wetlands can remove heavy metals and nutrients from run-off either through binding in sediments⁹³ or accumulation in plant tissues.⁹² Certain applications (such as heavy metal removal) require harvesting of plants in which metals are accumulated, providing a potential source of biomass for use as fuel.⁹⁴ However, there is evidence that preexisting microbial communities may have the ability to cleanse contaminated sites without interference in cases such as petroleum spills.⁹⁵ Recent advances in genetic engineering could lead to the application of such microorganisms in other situations.⁹⁶ A more complex issue within urban environments is the use of benthic sampling to monitor water or sediment quality, in which many systems of

habitat monitoring make use of ‘reference sites’. The difficulty in urban environments is that a wide range of aspects of biological functioning can be influenced by urban processes, making it difficult to tell whether, for example, water quality is driving the variation in biological communities. A solution to this problem would require either (1) the identification of new reference sites that are indicative of the types of water bodies found in urban environments but which are minimally impaired relative to other urban wetlands, or (2) the quantification of impact within a region such that comparisons are made internally and reflect *relative* quality rather than reference to an absolute standard.⁹⁷

Social Function

A further set of functions that differ from the direct and indirect uses described above relate to less tangible processes. Ponds certainly contain a greater proportion of the landscape-level biodiversity than other comparable habitats, as discussed above, and therefore allow humans to contribute to the conservation of endangered species that have specific habitat requirements.¹⁴ Urban ponds also contribute to green space in cities that may play a role in improving individual and community health and well-being.⁸⁹ Furthermore, there is a traditional and cultural link with wetlands that has been lost in many areas of the developed world but that is being increasingly encouraged as a focus in the conservation of wetlands.⁹⁰ This area in particular drives a need for interdisciplinary research incorporating both the physical/chemical/biological sciences and the social sciences in order to ensure that information about, and conservation of, wetlands is presented in such a way as to appeal to past or present cultural associations with those habitats.⁹⁰ Through a deeper understanding of the social function and value of wetlands and other ecosystems we can not only tailor our message to particular communities but also gain an insight into whether or how those cultural associations need to be shifted in order to ensure community buy-in to sustainable development.⁹¹

THE CHALLENGES OF URBAN BIODIVERSITY

Pollution

Partly as a result of their proximity to human activity and partly as a result of their deliberate use as filters of the waste from that activity, urban

wetlands tend to accumulate pollutants. Heavy metals also enter freshwaters through their association with deicing treatments. The application of road salt (predominantly NaCl, CaCl₂, and MgCl₂) as a treatment against ice is a common practice in many temperate, developed nations (for a review, see Ref 98). The salt itself causes problems for amphibian osmoregulation^{99,100} but also carries with it heavy metals that accumulate in the tissues of plants and animals (see Ref 101 for a review). While we understand the direct effects that salinization of urban wetlands can have on the constituent biota through the use of ecotoxicological assays and *in situ* biomonitoring, what is less clear are the indirect effects. For example, in an insightful mesocosm experiment, Meter et al.¹⁰² demonstrated that salt has an adverse direct effect on zooplankton, reducing competition for algae, which then increases the size and developmental rates of amphibians. Once more, this highlights the need to consider the ecosystem in its entirety, rather than focusing on a small number of species or components.

Invasive Species

Out of 891 species listed in the Global Invasive Species Database (<http://www.issg.org/database>), 277 (31%) are associated with urban areas, 395 (44%) are associated with water (wetlands, lakes, or water courses), and 147 are associated with both urban areas and water (16%, Figure 2). Urban wetlands provide two means for enhanced invasions. First, large numbers of species are imported into urban areas (either on purpose or by accident) creating a high propagule pressure that facilitates successful invasion.¹⁰³ Urban centers are nodes in trade and transport networks, which act as conduits and end-points for invasive species¹⁰⁴ and provide opportunities for a range of different introduction pathways.¹⁰⁵

Primary taxa that are imported deliberately into urban areas include ornamental species such as goldfish and aquatic plants, and pets such as terrapins and snakes. Particular issues arise when commercial suppliers of freshwater species misidentify what is being sold or fail to ensure adequate biocontrol measures are in place to prevent contamination with unordered organisms (such as molluscs and crustaceans).¹⁰⁶ The frequency of occurrence of aquarium species in shops¹⁰⁷ and the degree of importation of non-native species¹⁰⁸ have been shown to correlate with the likelihood of a species establishing in the wild for freshwater and marine fish, respectively. Garden plant communities can be extremely diverse¹⁰⁹ but these diverse communities

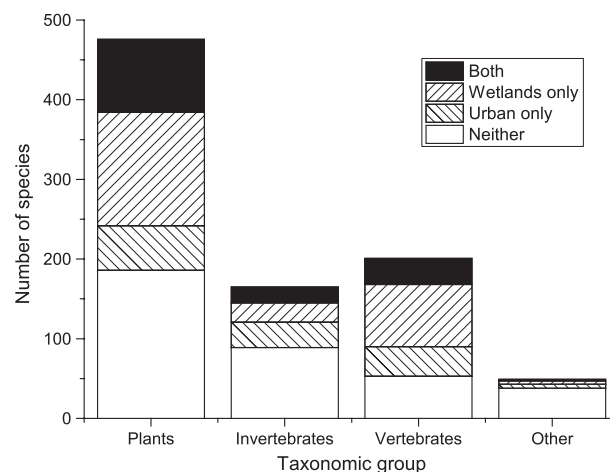


FIGURE 2 | Habitat use of 891 invasive species listed in the Global Invasive Species Database. 'Other' ($n = 49$) includes corals, fungi, microorganisms, oomycetes, sponges, and tunicates.

often comprise many ornamental, alien species (see above) and so act as a source for invasives to enter the surrounding natural or seminatural areas.^{110,111} Similarly, the occurrence of non-native fish in natural wetlands around urban centers increases with higher importation rates.¹¹² Deliberate human introductions of fish are also strongly implicated through the release of ornamental fish or sports fishing¹¹² and the increased chances of finding non-native species closer to human access points.⁵⁸

However, a second mode by which urbanization can facilitate invasive species is through the disturbance of existing ecosystems. Ehrenfeld¹¹³ provided a list of direct impacts from the presence of humans in or around aquatic environments, which can be summarized as follows: (1) modification of channels and banks; (2) disturbance from traffic; (3) presence of pet animals; (4) dumping of rubbish, and (5) reductions in permeability of surrounding land. However, as this list suggests, the impacts of urbanization are a heterogeneous in and of themselves. Rather than producing a single, invadable habitat type, human–wetland interactions in urban areas create a diversity of habitat types that sometimes favor invasive species.¹¹³

Pond Loss

Ponds have been lost across the developed world at a high rate as their function in agricultural and industrial settings declined and land was needed for urbanization or the intensification of agriculture. Much of this loss (50–70%) occurred during the late 19th and early 20th centuries, and what we witness now is a much-reduced rate of loss.¹¹⁴ Table 3 gives an international

TABLE 3 | An International Comparison of Patterns of Change in Pond Numbers

Country	Region	Dates	Pond Change (%)	Annual Change (%)	Primary Source	Cited in
UK (pre-1998)	Huddersfield	1985–1997	–31	–2.6	36	114
	North Leicestershire	1934–1979	–60	–1.33	115	114
	Bedfordshire	1910–1981	–82	–1.15	115	114
	Sussex	1977–1996	–21	–1.1	116	114
	London region	1870–1984	up to –90	–0.79	117	114
	Huntingdonshire	1890–1980	–56	–0.68	115	114
	Cheshire	1870–1993	–61	–0.5	118	114
	Essex (selected areas)	1870–1989	–55 to –69	–0.46 to –0.58	119	114
	Cambridgeshire	1840/90–1990	–68	–0.45 to –0.68	120	114
	Leicestershire	1840/90–1990	–60	–0.40 to –0.60	120	114
	Durham	1840/90–1990	–41	–0.27 to –0.41	120	114
	Clwyd	1840/90–1990	–32	–0.21 to –0.32	120	114
	Midlothian	1840/90–1990	–23	–0.15 to –0.23	120	114
	Edinburgh	1840/90–1990	–6	–0.04 to –0.06	120	114
	England and Wales	1880–1920	–57.5	–1.41	121	114
	Britain	1990–1996	–7.4	–1.23	122	114
	Great Britain	1900–1990	–75	–0.78	123	114
	Great Britain	1998–2007	+12.5	1.39	124	124
	England	1998–2007	+18.3	2.03	124	124
	Scotland	1998–2007	+5.5	0.6	124	124
	Wales	1998–2007	+16.9	1.88	124	124
Sweden		1914–1970	–55	–1.0	125	126
Netherlands		1900–1989	–90	–1.0	127	126
Denmark		1868–1974	–67	–0.6	Briggs (unpublished)	126
Germany	N Rhine Westphalia	1963–1986	> –40	> –1.7	128	126
	Berlin Sud	1880–1980	–81	–0.8	129	126
Poland	Wielopolska	1890–1941	–56	–1.1	130	126
Brazil		1905–2005	–90	–9.0	131	132

comparison of rates of pond loss, showing that such rates are similar across different countries and regions. Pond density is a key predictor of biodiversity,⁶⁵ which is unsurprising given the substantial barrier posed by the urban terrestrial matrix.⁵⁷ Pond loss, therefore, poses a far greater risk to the coherence of pond networks within urban environments than it does within other land-use contexts, and must be central to plans for the conservation of species in urban environments.⁶⁹ In some regions, such as the UK, new initiatives are seeking to create new ponds to replace those that are lost and those that are damaged, under the assumption that the restoration of degraded ponds is expensive and that those sites will never reach their former quality.¹³³ However, this hypothesis has received little empirical attention.

In urban environments, where space is limited, the restoration of existing ponds may be the only option. The key factor that will determine whether these pond creation initiatives can succeed is engagement with stakeholders such as the mineral extraction industry, developers, and the general public.

What is interesting is that the only study to have investigated changes in pond numbers in the last decade has shown a substantial increase in pond numbers from 1998 to 2007, mostly as part of the concerted effort to reverse the declines experienced in the UK over the past century mentioned above.¹²⁴ This pattern of change highlights the dynamic aspect of the pondscape, where ponds are created and lost at a much higher rate than is the case for other habitat types, which needs to be taken into account

when designing conservation measures. Conventional conservation practices tend to focus on particular sites, but a 'high-quality pond' at one time may not remain high quality (or even continue to exist) in the long term.¹⁹ There is also a lack of understanding about *how* to create a new pond. Observational studies based on preexisting sites suggest that the key requirements are as follows (summarized from the findings in Table 2):

- Avoid vertical walls that prevent amphibians from exiting the water,³⁹
- Maintain submerged and emergent plant communities⁴⁰ with light management,⁶⁰
- Situate to maximize connectivity with existing ponds,^{39,65,45}
- Be aware of human access that may influence species introductions,¹¹²
- Use for functions other than biodiversity, such as stormwater management, may not reduce diversity but may influence the composition of communities that occur,⁴⁴
- Plan a variety of pond types, as different conditions (e.g., fish/fishless, water chemistry, and morphology) support different communities in a wide range of taxa.^{54,40,43,66,62}

The field of freshwater conservation would benefit a great deal from experimental studies that investigate the accumulation and extent of biodiversity in networks of newly created ponds of varying kinds and configurations to establish best practice. Such studies should follow the example of Williams et al.¹³⁴ in planning, monitoring, and critically evaluating the success of pond network creation schemes, although methods would have to be adapted for urban environments.

ECOSYSTEM SERVICES AND CONFLICTING PRIORITIES

Biodiversity: The Good, the Bad, and the Ugly

In attempting to encourage aquatic biodiversity in urban areas, there is a need to consider the potential impacts of increasing nuisance species. Mosquitoes are a particular problem, and modifications of wetland design to reduce mosquito abundance (e.g., steepening of banks, removal of vegetation, and increased water depth¹³⁵) tend to remove those aspects of the habitat that promote other elements of biodiversity.¹⁷ Mosquitoes bring with them potentially fatal diseases

such as West Nile Virus (transmitted primarily by the *Culex pipiens-restuans* complex)¹³⁶ in North America and Dengue fever (transmitted by *Aedes aegypti*, a species adapted to urban environments, and *Aedes albopictus*) in southern Europe.¹³⁷ *A. albopictus* in particular is spreading through Europe,¹³⁸ and climate change may increase the latitude at which Dengue fever can be transmitted to produce seasonal waves of transmission in southern Europe.¹³⁹ However, mosquito production tends to be limited to a small number of sites, and so monitoring and management there can reduce the need to homogenize all urban wetlands.¹⁴⁰ Mosquitoes are also fewer in number when insect predators are present, providing an additional motivation for enhancing biodiversity in a broader sense.¹³⁶ Other nuisance species may be more innocuous, such as the loud calling of the striped marsh frog *Rana ridibunda*. Even for harmless animals, the general public holds deep-seated negative views (and even fear) of biodiversity¹⁴¹ and these views influence their preference for highly managed urban green space versus seminatural areas.¹⁴² However, such distaste can be overcome by gradual introduction of successively 'more-natural' sites coupled with outreach and education.¹⁴³

Aesthetics versus Ecology

Many surveys of residents show that there are generally positive views of urban wetlands.^{144–146} However, those wetlands are perceived as more attractive if they have highly visible mown areas and a clear view of the water without dense macrophyte beds,¹⁴⁷ suggesting that cultural sustainability needs to be considered alongside ecological integrity. The kind of management that seems to appeal most to residents also reduces biodiversity, for example, the removal of vegetation from city park ponds vastly reduces dragonfly richness.⁴⁰ The challenge, then, is to appeal to cultural sensitivities that may require ecological innovation,¹⁴⁸ or the provision of both 'natural' and 'formal' green spaces from which the public extract different benefits.¹⁴⁹

OPPORTUNITIES FOR THE FUTURE

Green Space

Increasing urbanization involves not only the sprawl of urban margins but also the intensification of land use within already built-up areas and this reduction in land per unit population leads to a concomitant reduction in available green space. It is generally accepted that green space has a positive effect on a range of health outcomes.⁸⁹

In particular, biodiversity seems to be strongly associated with psychological benefits.¹⁵⁵ However, the construction of new developments necessitates the construction of drainage that can provide wetland habitats,¹⁵⁶ and these wetlands can contain considerable biodiversity.^{53,157,158} The creation of wetlands within such developments can be guided by ecological theory, but this will require ecologists to turn our observations of variation in biodiversity in artificial, urban wetlands, e.g., Refs 53 and 158–160, into guidance for developers. Further opportunities may lie in the green buildings themselves, into the walls and roofs of which water features can be integrated.

Garden Ponds

On the basis of an analysis of multiple estimates of pond prevalence in the UK gardens, Davies et al.³³ suggest that around 16% (95% CI: 0.11–0.20) of gardens contain ponds. The authors extrapolate from these surveys to give a predicted garden pond resource of 2.5–4.5 million ponds across the country, with an estimated surface area of 3.5 km². The difficulty with monitoring this resource is the size and the lack of detailed mapping data: estimates of mean garden pond size vary from 1 m²³³ to 2.5 m².³⁴ and these are not shown on maps. Furthermore, difficulty with access for researchers to garden ponds may have deterred earlier work.¹¹⁴ However, garden ponds are well used by amphibians, which show little habitat preference but may be influenced by the presence of fish.¹⁶¹ Attempts at experimental supplementation of garden ponds using small mesocosms (0.21 m²) suggest that a range of animals may readily and rapidly colonize even these small wetlands, and that there could be an additional value as supplementary habitat for amphibians.³⁴ This resource must be incorporated into urban ecology in a more comprehensive way in order to adequately evaluate aquatic ecological processes.

Research

Urban wetlands provide a replicated set of habitats that are close to centers of research as well as large numbers of lay people. This opens the way for citizen science, a growing area with many innovative and extensive research projects.¹⁶² Particular ecological questions that can be asked include investigations of metapopulation¹⁶³ and metacommunity¹⁶⁴ ecology within the context of network theory.⁷⁹ The harsh environmental pressures exerted on urban ecosystems also provide opportunities to study evolutionary processes.¹⁶⁵ What effect does acute and chronic exposure to road salt pollution have on invertebrates and amphibians? Can low matrix permeability

facilitate speciation in urban versus rural populations? How do mutualists adapt when urbanization removes one component of the interaction? The developing world, where systems are far closer to natural prior to urbanization, would give a clearer picture of the impacts of urbanization than the space-for-time urban–rural gradient studies that are conducted in Europe and North America, e.g., Ref 166.

In parallel to the opportunities presented for fundamental research, applied research must also be directed toward the needs of end-users. These include hydrologists who use stormwater facilities to manage surface flow—while some studies show evidence of successful use of wetlands to reduce flooding,¹⁶⁷ smaller scale interventions have been little studied. As discussed above, conservation agencies require evidence for the efficacy of particular designs and configurations of ponds in order to maximize benefits from limited resources. Interest in nutrient and pollutant retention has also tended to rest on observational studies, while studies of the efficacy of different methods of pollutant retention and of the impacts of urban pollutants are required.¹⁶⁸

CONCLUSION

The biodiversity resource represented by urban ponds is currently poorly quantified and described. The majority of urban ponds are likely to be located in residents' gardens and represent a habitat that is almost completely unknown, although the high-quality urban ponds are likely to be larger, more diverse habitats managed as nature reserves or stormwater management facilities. A biotic homogenization of biological communities seems to be a useful concept within which to consider deliberate modifications of the environment, although it seems that there is a great deal of variation in the extent to which homogenization occurs. The few studies of habitats such as stormwater ponds and industrial ponds highlight the diversity of factors affecting biodiversity and the impact of a wetland's past use on its future management. Given the increasing rate of urbanization, particularly in the developing world, a better understanding of urban ecosystems is essential to the protection of biodiversity in general.

The interaction between humans and urban freshwater ecosystems stands as a representative case study for the interplay between natural and anthropological processes. I have tried to illustrate some of these in Figure 3. First, it is becoming increasingly clear that the desire for ecosystem services and development can both increase and decrease freshwater resources along with their constituent

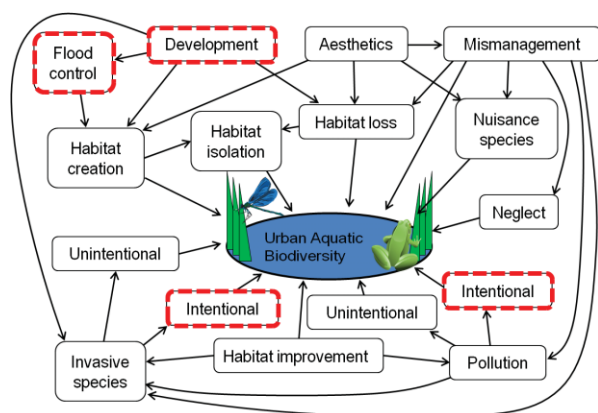


FIGURE 3 | Multiple interacting factors affecting urban aquatic biodiversity. Red, dotted boxes indicate areas of legal influence.

biodiversity. This phenomenon serves to emphasize the opportunity for sustainable urbanization and incorporation of diverse wetlands, but this kind of complementary consideration of biodiversity and other ecosystem services requires effective, interdisciplinary, socio-ecosystem approaches. The mismanagement of the existing freshwater resource can cause or exacerbate a range of problems, often owing to limited evidence base for the management of urban wetlands. There is a great need to study further the interactions between natural and artificial water bodies within urban contexts to be able to advise land managers and the general public about best practice for urban freshwater management. In addition, better communication of the damage done by, for example, invasive species would not only benefit native biodiversity but also reduce the costs of dealing with nuisance invasive species once they are established. Finally, on the ecological interactions between ponds, it is clear that simple gain and loss of particular habitats will be amplified through the effects on isolation (or, conversely, connectivity).

Particularly exciting are the opportunities to engage the general public (particularly school children) in biodiversity conservation efforts in an attempt to bring about broader support for environmental protection through increased familiarity with nature (Box 2). Evidence-based engagement, designed to bring the public into contact with nature

without evoking negative reactions, will be necessary. More broadly, urban wetlands bring a range of scientific phenomena to the taxpayer's doorstep, including bioremediation, ecology, conservation, chemistry, hydrology, and climate change research. Indeed, for each opportunity that wetlands present for researchers, there is an accompanying opportunity for public engagement in the process and outcome of that scientific research. Taking advantage of these opportunities, proactive engagement with the public, and innovation to accommodate competing ecosystem services will pay dividends in terms of accelerated progress toward sustainability.

BOX 2

EDUCATIONAL VALUE OF URBAN PONDS

It has been proposed that the key to future conservation of wildlife is through increasing the exposure of the general public (whose tax money will fund much of the conservation work) to nature in urban areas.¹⁵⁰ Aside from the substantial value of urban biodiversity to the functioning of urban ecological systems, there is great benefit to be obtained from the provision and use of urban wetlands in public education. These applications are straightforward in subjects such as the sciences,¹⁵¹ but are easily adapted to other parts of the curriculum, and health and safety considerations are laid out elsewhere.¹⁵² Some have suggested that negative perceptions of wetlands are not innate but learnt through childhood, and so exposure to ponds in a positive light may enhance the next generation's perceptions of water.¹⁵³ This may be appropriate not only in schools, where small wetlands can be used as teaching resources, but also around community centers and nature reserves with access to the general public. Indeed, any time that a wetland is created or retrofitted there exist opportunities to promote recreation and education through the integration of interpretative signage and public access.¹⁵⁴

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